



Comparing multiple representations to calculate theoretical probabilities for combined events

1 The probability that an event will occur is the proportion of times the event is expected to happen in a suitably large experiment. Fill in the blank

2 A trial has three possible outcomes: {A, B, C}. $P(A) = \frac{2}{7}$. $P(B) = \frac{1}{7}$. $P(C) = \frac{4}{7}$. Which outcome is most likely to happen? Tick **1** correct answer

☐ A

☐ B

☒ C

3 There are two trials. In Trial 1, a spinner with {A, B, C} is spun twice. In Trial 2, a spinner with {B, C} is spun twice. Which statement is true? Tick **1** correct answer

☒ The likelihood of A or C appearing in the first trial is the same.

☐ The likelihood of A appearing in the second trial is high.

☐ The likelihood of B appearing in both trials is more than equal chance.

4 In a fair six-sided die, what is the probability of rolling a 9 and getting 'tails' in a single toss of a fair coin? Fill in the blank

0

5 A coin is flipped twice. Complete the sample space $\xi = \{HH, HT, \underline{\hspace{2cm}}, TT\}$ Tick **1** correct answer

☐ HH

☐ HT

☒ TH

☐ TT



6 When rolling a fair six sided die, which event is more likely to occur? Tick **1** correct answer



Rolling an even number



Rolling a prime number > 2



Rolling a multiple of 3

